

QUESTION 1

What is the value of $154_8 + 132_8$?

- A. 286_8 B. 22_8 C. 386_8 D. 306_8 E. 3217_8

QUESTION 2

What is the value of `z` after executing the code to the right?

- A. 10 B. 221 C. 1491
D. 58 E. 280

```
int x = 10;
int y = x * 3 + 7;
int z = y + 3 * 7;
```

QUESTION 3

What are the values of `ch` and `total` after executing the code to the right?

- A. `ch: 'z'` B. `ch: 'z'`
 `total: 26` `total: 25`
C. `ch: 'x'` D. `ch: 'x'`
 `total: 25` `total: 24`
E. `ch: 'y'`
 `total: 24`

```
int total = 0;
char ch = 'a';

while (ch < 'y') {
    total++;
    ch++;
}
```

QUESTION 4

Which of these builds a `MyClass` object with data member `x` initialized to 10 and data member `y` initialized to 0.5?

- A. `MyClass m = (10, 0.5)`
B. `MyClass m = {10, 0.5}`
C. `MyClass m(10, 0.5)`
D. `MyClass(10, 0.5)`
E. `new MyClass(10, 0.5)`

```
public class MyClass {
    public MyClass(int xInit, double yInit) {
        x = xInit;
        y = yInit;
    }

    public int getX() { return x; }
    public double getY() { return y; }

    private int x;
    private double y;
}
```

QUESTION 5

Which of these can be used inside any class to output data member `y` from `MyClass m`?

- A. `System.out.print(m(y))`
B. `System.out.print(m.getY())`
C. `System.out.print(m/y)`
D. `System.out.print(m.y)`
E. Both B and D

QUESTION 6

Which of these adds 1 to `int x` and returns the new value?

- A. `x++` B. `++x` C. `x + 1` D. `1 + x` E. Both A and C

QUESTION 7

What is output by the code to the right?

- A. r B. o C. l
D. orld E. nothing

```
String s = "hello world";
System.out.print(s.charAt(8));
```

QUESTION 8

What replaces **<*1>** in the code to the right to give the number of characters in `String s`?

- A. `s.numChars()` B. `s.length()`
C. `s.numChars` D. `s.length`
E. `s.size`

```
public static int compute(String s) {
    int total = 0;
    for (int i=0; i < <*1>; ++i) {
        char c = s.charAt(i);
        if (c >= '0' && c <= '9')
            total = total + (c - '0');
    }
    return total;
}
```

QUESTION 9

Assume **<*1>** is filled in correctly. What is returned by `compute("2006")`?

- A. 0 B. 8 C. 10
D. 12 E. 2006

QUESTION 10

What is the value of `total` after executing the code to the right?

- A. -2 B. -1 C. 0
D. 1 E. 2

```
int x = 42, total = 0;
while (x != 0) {
    switch(x%3) {
        case 0: total = total + 1; break;
        case 2: total = total - 1; break;
    }
    x = x/3;
}
```

QUESTION 11

What is returned by `process(5)`?

- A. "ababa" B. "babab"
C. "aaaaa" D. "bbbbbb"
E. "bbbbbb "

```
public static String process(int x) {
    if (x == 0)
        return "";
    else if (x<0)
        return "a" + process(-x-1);
    else
        return "b" + process(-x+1);
}
```

QUESTION 12

How many a's are in the string returned by `process(1001)`?

- A. 0 B. 500 C. 501
D. 1000 E. 1001

QUESTION 13

Can these static methods all be part of the same class?

- A. Yes
- B. No, because the names of the methods are the same
- C. No, because the first and second method have the same name, return type, and parameter names
- D. No, because the methods have the same name and they all have two parameters
- E. No, because the first and third method have the same name and parameter types

```
public static int f(int x, int y) {
    // code not shown
}

public static int f(int x, double y) {
    // code not shown
}

public static double f(int a, int b) {
    // code not shown
}
```

QUESTION 14

Which of these replaces **<*1>** in the code to the right to create an array of the number of the days in each month that is accessible only inside the `Date` class? (Use 29 for February.) There should be only one copy of the array, not one for each instance of `Date`.

- A. `int[12] daysPerMonth = {31, 29, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31};`
- B. `private int[] daysPerMonth = {31, 29, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31};`
- C. `private static int[] daysPerMonth = {31, 29, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31};`
- D. `static int[12] daysPerMonth = {31, 29, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31};`
- E. More than one of these

```
public class Date {
    public Date(int day, int month, int year)
    {
        if (year <= 0)
            throw new IllegalArgumentException(
                "Bad year");
        if (month <= 0 || month > 12)
            throw new IllegalArgumentException(
                "Bad month");
        if (day <= 0 ||
            day > daysPerMonth[month-1])
            throw new IllegalArgumentException(
                "Bad day");
        if (month == 2 && day == 29 &&
            !isLeapYear(year))
            throw new IllegalArgumentException(
                "Bad day");
        this.day = day;
        this.month = month;
        this.year = year;
    }

    private static boolean isLeapYear(int y)
    {
        if (y%400 == 0) return true;
        if (y%100 == 0) return false;
        if (y%4 == 0) return true;
        return false;
    }
}
```

QUESTION 15

Assume **<*1>** is filled in correctly. Which of these causes an `IllegalArgumentException` to be thrown?

- A. `new Date(31, 7, 2000)`
- B. `new Date(31, 7, 2100)`
- C. `new Date(29, 2, 2000)`
- D. `new Date(29, 2, 2100)`
- E. More than one of these

```
// other methods not shown

private int day, month, year;
<*1>
}
```

QUESTION 16

What replaces **<*1>** in the code to the right to return whether `elements` is empty?

- A. `elements.length() == 0`
- B. `elements.size() == 0`
- C. `elements.number() = 0`
- D. `elements == null`
- E. `elements[0] == null`

For the remaining questions, assume that **<*1>** has been filled in correctly.

QUESTION 17

What is output by static method `test()`?

- A. 312 B. 321 C. 231
- D. 123 E. 132

QUESTION 18

What is the running time of the `enqueue()` method when the queue contains n items? Choose the most restrictive correct answer.

- A. $O(1)$ B. $O(\log n)$ C. $O(n)$
- D. $O(n^2)$ E. $O(n \log n)$

QUESTION 19

On which of these arrays would static method `process()` return 3 if the array was passed as its parameter?

- A.

1	2	3	4	5	6
---	---	---	---	---	---
- B.

6	5	4	3	2	1
---	---	---	---	---	---
- C.

1	3	5	2	4	6
---	---	---	---	---	---
- D.

2	4	6	1	3	5
---	---	---	---	---	---
- E.

1	3	5	6	4	2
---	---	---	---	---	---

```
public class MyQueue<E> {

    public MyQueue() {
        elements = new LinkedList<E>();
    }

    public void enqueue(E element) {
        elements.addLast(element);
    }

    public E dequeue() {
        return elements.removeFirst();
    }

    public boolean isEmpty() {
        return <*1>;
    }

    private LinkedList<E> elements;

    public static void test() {
        MyQueue<Integer> testQ =
            new MyQueue<Integer>();
        testQ.enqueue(1);
        testQ.enqueue(2);
        testQ.enqueue(3);
        System.out.print(testQ.dequeue());
        System.out.print(testQ.dequeue());
        System.out.print(testQ.dequeue());
    }

}
```

```
public static int process(int[] array) {
    int returnValue = 0;
    for (int i=0; i<array.length-1; ++i)
        if (array[i]<array[i+1])
            ++returnValue;
    return returnValue;
}
```

QUESTION 20

What replaces <*1> and <*2> in the code to the right so that it is syntactically correct?

- A. <*1>: abstract class
<*2>: extends
- B. <*1>: interface
<*2>: extends
- C. <*1>: abstract class
<*2>: implements
- D. <*1>: interface
<*2>: implements
- E. Either A or D

```
public <*1> Shape {
    public abstract double area();
    public abstract double perimeter();
}

public class Rectangle <*2> Shape {
    public Rectangle(int length, int width) {
        l = length;
        w = width;
    }
    public double area() {
        return l * w;
    }
    public double perimeter() {
        return (l + w) * 2;
    }
    private int l, w;
}
```

QUESTION 21

Assume <*1> and <*2> are filled in correctly. Which of these can be used in any class to output the perimeter of Rectangle r?

- A. System.out.print(2*(r.l + r.w))
- B. System.out.print((r.l + r.w)*2)
- C. System.out.print(r.perimeter())
- D. System.out.print(perimeter(r.l,r.w))
- E. All of these

QUESTION 22

What is output by the code to the right?

- A. hellox B. helloxz
- C. hellooz D. helloox
- E. The code contains an infinite loop

```
String s = "hello";
int i = 0;
try {
    while (true) {
        System.out.print(s.charAt(i));
        ++i;
    }
}
catch(Exception e) {
    System.out.print('x');
}
finally {
    System.out.print('z');
}
```

QUESTION 23

What is output by the code to the right?

- A. 89 B. 88
- C. 98 D. 18
- E. 9

```
int x = 2^3;
int y = (int)Math.pow(2.0,3.0);

System.out.print("" + x + y);
```

QUESTION 24

Which of these input strings would cause the code to the right to set x to 10, y to 0.5, and s to " 16"?

- A. 100.516 B. 10 0.5 16
- C. 10.5 16 D. 10.516
- E. Both B and C

```
Scanner input = new Scanner(System.in);
int x = input.nextInt();
double y = input.nextDouble();
String s = input.nextLine();
```

QUESTION 25

What is output by the code to the right?

- A. hatfalse B. hattrue
- C. thatfalse D. thattrue
- E. hfalse

```
String s = "hat", t = "that";
System.out.print(t.substring(1));
System.out.print(s == t.substring(1));
```

QUESTION 26

What replaces **<*1>** to call the MySuperClass constructor with the parameter 0?

- A. super; B. super();
- C. this(0); D. super(0);
- E. MySuperClass(0);

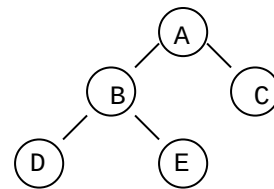
```
public class MySubClass
    extends MySuperClass {
    public MySubClass() {
        <*1>
    }

    // other data and methods not shown
}
```

QUESTION 27

What order are the nodes processed in a preorder traversal of this binary tree?

- A. A B C D E B. E D C B A
- C. D E B C A D. E D A C B
- E. A B D E C

**QUESTION 28**

What is returned by m.entrySet()?

- A. A Set containing Map.Entry objects
- B. A Set containing Integer objects
- C. A Set containing String objects
- D. An Iterator
- E. A Map from Integer objects to String objects

```
Map<Integer, String> m =
    new TreeMap<Integer, String>();
```

QUESTION 29

Which of these best describes the static method `sort()` to the right?

- A. Sorts in ascending order by finding the smallest remaining item each pass of the outer loop
- B. Sorts in ascending order by finding the largest remaining item each pass of the outer loop
- C. Sorts in descending order by finding the smallest remaining item each pass of the outer loop
- D. Sorts in descending order by finding the largest remaining item each pass of the outer loop
- E. Does not modify the array

```
public static void sort(Comparable[] array)
{
    for (int i=0; i<array.length-1; ++i) {
        Comparable c = array[i];
        int cIndex = i;
        for (int j=i+1; j<array.length; ++j)
            if (c.compareTo(array[j])<0) {
                c = array[j];
                cIndex = j;
            }
        array[cIndex] = array[i];
        array[i] = c;
    }
}
```

QUESTION 30

What is the running time of `sort()` on an array containing n items? Choose the most restrictive correct answer.

- A. $O(1)$
- B. $O(\log n)$
- C. $O(n)$
- D. $O(n^2)$
- E. $O(n \log n)$

QUESTION 31

What replaces `<*1>` in the code to the right to cause all of the items in `intArray` to be output?

- A. `for (item in intArray)`
- B. `for (item : intArray)`
- C. `for (int item : intArray)`
- D. `for (int item : int [] intArray)`
- E. `for (int item in intArray)`

```
int [] intArray;
// code to initialize intArray not shown

<*1> {
    System.out.println(item);
}
```

QUESTION 32

What replaces `<*1>` in the code to the right to give the character at position `i` in `s`, changing it to upper case if it is lower case?

- A. `Character.toUpperCase(s.charAt(i))`
- B. `Character.toUpperCase(s.charAt(i))`
- C. `Character.isLowerCase(s.charAt(i))`
- D. `s.charAt(i) + 'A' - 'a'`
- E. `s.charAt(i) + isLowerCase?32:0`

```
public static String up(String s) {
    StringBuffer sb = new StringBuffer(s);
    int i = 0;
    while (i < s.length()) {
        sb.setCharAt(i, <*1>);
        ++i;
    }
    return sb.toString();
}
```


QUESTION 37

What replaces `<*1>` in the code to the right to return true when the ratio of worth to square feet is higher than the ratio for the other Deed? Use floating point arithmetic for the division.

- A. `((double)worth)/squareFeet > ((double)d.worth)/d.squareFeet`
- B. `(double)(worth/squareFeet) > (double)(d.worth/d.squareFeet)`
- C. `(double)(worth)/squareFeet > (double)(d.worth)/d.squareFeet`
- D. `double(worth)/squareFeet > double(d.worth)/d.squareFeet`
- E. Both A and C

```
public class Deed {
    // constructor and some methods not shown

    public boolean goodValue(Deed d) {
        return <*1>;
    }

    private String address;
    private int squareFeet;
    private long worth;
}
```

QUESTION 38

What is the super class of Deed?

- A. Object
- B. null
- C. Deed
- D. super
- E. It has no super class

QUESTION 39

On which of these arrays does the static method call `search(array, 6)` return 4?

- A.

1	3	5	5	5	7
---	---	---	---	---	---
- B.

2	3	4	5	6	7
---	---	---	---	---	---
- C.

18	3	24	12	6	10
----	---	----	----	---	----
- D. Both A and B
- E. Both B and C

```
public static int search(int[] array,
                        int item) {
    int min=0, max = array.length, mid=0;
    while (min < max) {
        mid = (min + max)/2;
        if (array[mid]==item) break;
        if (array[mid]<item)
            min = mid+1;
        else max = mid;
    }
    return mid;
}
```

QUESTION 40

Which of these is the binary equivalent of the hexadecimal number 37_{16} ?

- A. 1111111_2
- B. 101111_2
- C. 110111_2
- D. 111011_2
- E. 111101_2

Computer Science Answer Key

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1. D	11. B	21. C	31. C
2. D	12. B	22. B	32. A
3. E	13. E	23. D	33. D
4. E	14. C	24. B	34. A
5. B	15. D	25. A	35. E
6. B	16. B	26. D	36. D
7. A	17. D	27. E	37. E
8. B	18. A	28. A	38. A
9. B	19. E	29. D	39. D
10. C	20. E	30. D	40. C